Project Initialization and Planning Phase

Date	03 July 2026
Project Title	**Credit Card Approval Prediction**
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to **automate the credit card approval process** using Machine Learning, improving the speed, accuracy, and consistency of credit decisions. By replacing manual evaluation with an intelligent prediction system, the solution helps financial institutions reduce processing time, minimize human errors, identify high-risk applicants, and enhance customer satisfaction. The project integrates a trained Machine Learning model with a **Flask web application** and supports deployment through **IBM Watson Machine Learning** for real-time cloud-based predictions.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based Credit Card Approval Prediction System that enables faster, more accurate, and reliable approval decisions by analyzing applicant financial and demographic information.
Scope	The project includes data preprocessing, feature engineering, model training using multiple classification algorithms, model evaluation, Flask web application development, and cloud deployment using IBM Watson Machine Learning for real-time predictions.
Problem Statement	
Description	Banks receive thousands of credit card applications daily. Manual evaluation is time-consuming, inconsistent, and susceptible to human error. Approval decisions depend on multiple financial and personal factors, making the process complex and resource-intensive.
Impact	Automating the approval process improves operational efficiency, reduces decision-making time, minimizes financial risk, enhances prediction accuracy, and provides customers with faster eligibility assessments.
Proposed Solution	
Approach	Develop a Machine Learning model that analyzes applicant information to predict credit card approval or rejection. The best-performing model is integrated into a Flask web application and deployed on IBM Watson Machine Learning for cloud-based predictions.

Key Features	• Machine Learning-based credit card approval prediction. • Automated applicant eligibility assessment. • Real-time prediction through a Flask web application. • Comparison of multiple ML algorithms (Logistic Regression, Decision Tree, Random Forest, XGBoost). • Cloud deployment using IBM Watson Machine Learning. • Scalable and user-friendly interface.

Resource Requirements:

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Computing Resources	Intel Core i3/i5 or above (GPU optional for training)
Memory	Memory	8 GB RAM
Storage	Disk space for data, models, and logs	20 GB free disk space (SSD recommended)
Software		
Frameworks	Framework	Flask
Libraries	libraries	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	Development Environment	Python 3.10+, Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source	Kaggle / UCI Machine Learning Repository
	Dataset	Credit Card Approval Dataset
	Format	CSV
	Size	Approximately 690 records (UCI) or larger Kaggle datasets